MAMA: MATERNAL ANALYTICS AND MONITORING ALLIANCE

ISSUE

1. Maternal and infant mortality is of great concern in Sub-Saharan Africa, as the lesser-developed countries in the African continent have the highest maternal and infant mortality rates in the world. Despite improvements made in the last two decades since 2000, the mortality rate of both women and children under 5 is still on an uprise in Africa compared to its neighbouring continents. (Figure 1.1, 1.2) This is the result of a gap in adequate maternal healthcare and education, despite the fact that women of African descent are susceptible to pre-existing cardiovascular morbidity that increases the risk of maternal mortality, and to adverse pregnancy outcomes that put them at risk of developing long-term cardiovascular disease.[1] This has created a concern for international, and local organizations, as it has demonstrated the need for the improved pursuance of SDG 3. This brief will explain pressing issues and developments, and will make recommendations - specifically to improve the UN's strategies to harness modern technology and ensure that each expecting mother and her child are prioritized in the early stages of her pregnancy, up until the delivery and early years of her baby - specifically in Sub-Saharan Africa.

BACKGROUND | Reference here for our background.

1. In 2024, sub Saharan Africa experienced 442 maternal deaths per 100 000 and 51 infant deaths per 1,000.[2] The issue of poor pre and postnatal care is rooted in the region's colonial history. Colonial regimes built healthcare systems that served settlers while excluding the majority of the population, especially in rural areas. This created lasting gaps in access to maternal and neonatal care, including skilled birth attendants and emergency services. Today, many women still face structural barriers to safe childbirth, and infants often die from preventable causes due to weak health infrastructure.[3] These outcomes are not just medical failures, but enduring consequences of colonial neglect and systemic inequality.[4] (Figure 1.3, 1.4)

ALTERNATIVE OPTION AND ANALYSIS

2. **Quarterly Health Literacy Training:** Integrate health literacy programming into primary and secondary school curriculum, co-devised by UN Women and local school systems. Here, students will engage in interactive learning in child nutrition, pregnancy and prenatal care, and water sanitation. Recognizing the cultural sensitivities surrounding the Western model of sex education that may be taboo in some communities, the program will emphasize practical knowledge in hygiene, nutrition, and caregiving for pregnant women and children. This will equip youth with knowledge to prevent adverse health outcomes and support long-term well-being, rather than attempt to restructure *concrete* social norms.[5] (Figure 1.5, 1.6).
   a. *Analysis:* A comprehensive health literacy approach would empower communities through preventative measures that address adverse health effects associated with vulnerable mothers and infants. The quarter-annual nature of training would ensure retention and consistent learning, especially in areas where maternal and infant mortality is highest, and the average age of pregnancy is youngest.

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[1] Lister, Rolanda L, et al. "Black Maternal Mortality - the Elephant in the Room." *World Journal of Gynecology & Womens Health*, vol. 3, no. 1, 22 Nov. 2019, www.ncbi.nlm.nih.gov/pmc/articles/PMC7384760/, https://doi.org/10.33552/wjgwh.2019.03.000555. Accessed 3 May. 2025.
[2] "African Region's Maternal and Newborn Mortality Declining, but Progress Still Slow." *WHO | Regional Office for Africa*, 2025, www.afro.who.int/news/african-regions-maternal-and-newborn-mortality-declining-progress-still-slow.
[3] "Sepsis and Malnutrition Stalk the New Mothers and Babies of South Darfur." *AP News*, 24 Sept. 2024, www.apnews.com/article/ed8235d3b11a4a6ea2fcf27b7013d20c. Accessed 3 May. 2025.
[4] Cueto, Marcos. "A History of Global Health: Interventions into the Lives of Other Peoples by Randall M. Packard." *Bulletin of the History of Medicine*, vol. 91, no. 4, 2017, pp. 816–817, https://doi.org/10.1353/bhm.2017.0094. Accessed 3 May. 2025.
[5] Amanu A, Adamu, et al. *Risk Management and Healthcare Policy*, vol. Volume 16, Mar. 2023, pp. 425–437, https://doi.org/10.2147/rmhp.s399196.

APPROACH

3. To narrow our understanding of the disparities faced in Sub-Saharan Africa with regards to its mortality rate, we used the UNStats SDG Database to gather data on global Maternal and Infant Mortality, and the Human Fertility Dataset to gather data on the Average Age of First Child. We then utilized the WITS Dataset to gather data on GDP per Capita by Year, Python for data processing and EDA, and then Tableau for Data Visualization and Dashboards - all of which to plot the geographic incidence of mortality.

RECOMMENDATION

4. **Maternal Analytics & Monitoring Alliance (MAMA):** We will create a Smart Birth Registry and Maternal Health Watchlist, where each pregnant woman who accesses a UNICEF mobile clinic or UNW clinic based on our findings of which areas have the highest incidence, will be automatically enrolled in our centralized digital health monitoring system and registry. This will track both the woman and child's vaccinations, medical visits, and health milestones from pregnancy to age 5. Using machine learning, we will develop a real-time watchlist to identify each mother at risk of severe complications based on their location and medical history, and use large-scale predictions to determine which areas require the most attention in maternal and infant healthcare to deliver accessible care. (Figure 1.7) We will use these large-scale predictions to work with local NGOs (e.g Coalition for Women's Health in Africa) to improve service. We will provide customized reminders and digital care services to pregnant women and mothers. Our model will use SMS/WhatsApp/mobile app notifications for vaccinations, breastfeeding and weaning milestones, growth, vitamin supplementation, illnesses, nutritional tips relevant to the child's age and location, and emergency visits. We will provide information on nearby health camps, food distribution drives, and mobile clinics.

   a. **Ethical Considerations:**
      i. *Culturally Sensitive Protocols:* UN Women will provide consent forms for those who choose to opt in to MAMA, in addition to clear communications on the objectives of MAMA in local languages and dialects, using visuals or verbal explanations for those with limited literacy.
      ii. *Voluntary Participation:* Make it clear that enrollment is voluntary and healthcare will not be denied if someone opts out. Moreover, mothers will be allowed to choose what data they're comfortable sharing (e.g. basic health monitoring vs. full participation in predictive analytics). Through anonymization, we will ensure that personal information is anonymized for predictive analytics. Only authorized healthcare workers and UN analysts should access sensitive data.
      iii. *Ethics Review Board:* Community leaders and healthcare professionals shall review complaints on the ethical delivery of healthcare tracking and services. They will regularly consult with local communities and maternal health professionals to review the system and adjust as needed.

   b. **Analysis:**
      i. *(+) Accessible Care & Public Health Savings:* Harnessing technology and predictive analytics to provide accessible, personalized care can identify high-risk pregnancies earlier, preventing complications and reducing mortality. It can improve accessibility and timely interventions, reduce costly emergency responses, promote healthier early childhood development, and decrease the weight of demands on future health services. Data-driven planning can streamline and maximize efficient allocation of existing resources (i.e mobile clinics, staff, supplies). (Figure 1.7)
      ii. *(-) Initial Setup and Operations:* It may be expensive in terms of technology and manpower to afford the machine learning technology necessary to pursue predictive analytics, whilst also maintaining system updates, consent protocols and feedback mechanisms.

APPENDIX

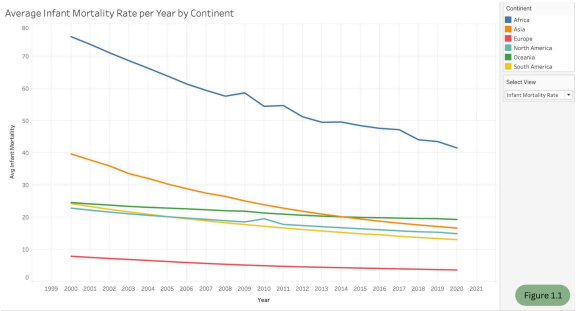


Figure 1.1

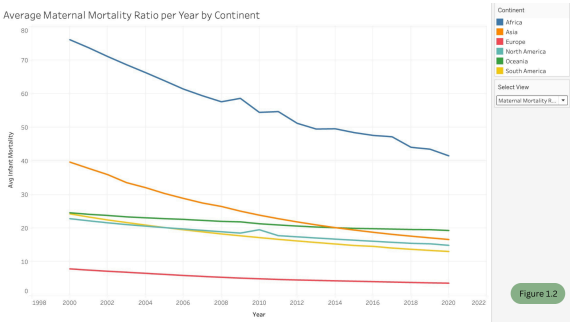


Figure 1.2

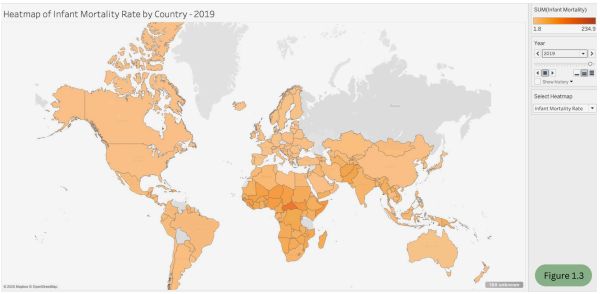


Figure 1.3

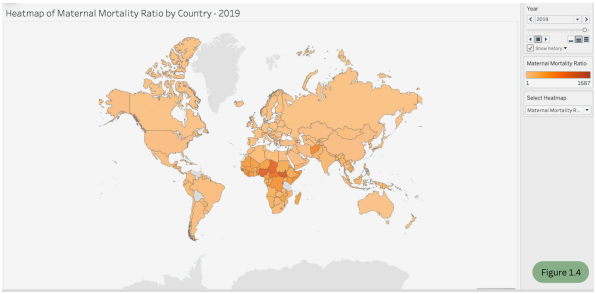
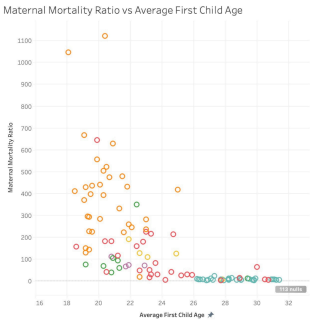


Figure 1.4



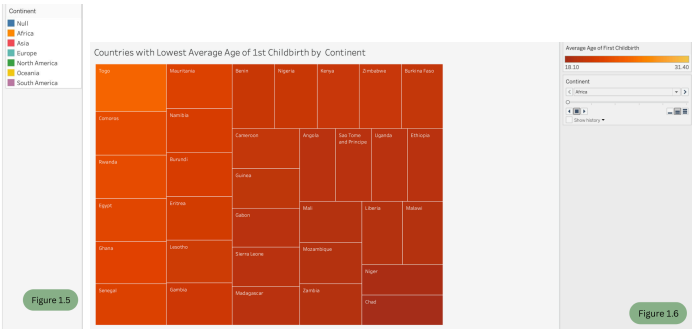


Figure 1.5

Figure 1.6

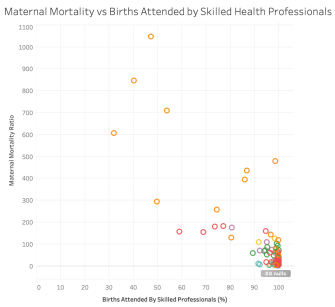


Figure 1.7